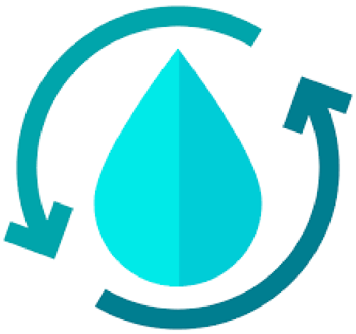
Sustainable Marketing Project

Xian Zhou (V00929234)

WaterWise

Overview

The service we've envisioned focuses on retrofitting older businesses within the CRD with greywater collection systems for their washrooms. The process would involve diverting



water from sink drains and rainwater and collecting it in a tank. Depending on the business, there may be other sources of water to collect. From the tank receptacle, the water would be cleaned through a series of filters. Then, that filtered water would be used to flush the toilets. The purpose is to reduce the amount of freshwater the businesses use, providing savings on utility bills and helping to conserve water supplies. This cost-effective, environmentally friendly solution to water conservation solves the problem of water wastage by recycling and reusing the water that would otherwise discharge into the sewer system.

Sustainable Development Goals

The backbone of our service is based on sustainable goals that have been highlighted by the United Nations. We've touched on many of these goals, first being Responsible Consumption and Production, in our case responsible consumption of valuable and limited freshwater. Our goal is to change one building and company at a time to slowly transition to an overall more resourceful and sustainable community. This service will help to develop the technology of greywater collection and make it used city-wide, and eventually across the country. This speaks towards the UN's Sustainable Development Goals of Sustainable Cities and Communities and Industry, Innovation, and Infrastructure. We also promote the UN SDG goal of Clean Water and

Sanitation. By recycling greywater and reusing it in washrooms, businesses will be overall reducing their water usage as well as saving the freshwater for its more valuable uses, such as for drinking and hand washing.

Research

<u>Source 1</u>

UVic Sustainable Action Plan (*Sustainability Action Plan: Campus Operations 2014-2019*, n.d.)

The facilities department of University of Victoria continues to provide sustainability services, especially on energy & water management. Staff work to reduce energy and water consumption by monitoring and analyzing data. University of Victoria has set up and kept operating a Sustainability Action Plan of campus operation from 2014-2019. The specific goal was to reduce campus water consumption by 25%. Reducing water consumption by 25% means there would be only ¾ of the utility fee used compared to the fee before the plan. The data indicates that UVic has great intentions on achieving water sustainability back in 10 years.

From the UVic webpage of Campus Planning and Sustainability, the updated 2020-2021 Sustainability Action Plan indicates the campus water reduction strategies. One of the strategies is to reduce irrigation during summer months. That raises the question of whether reducing irrigation during summer months would disrupt the ecological balance of the campus. Therefore, the gray water collection system will work effectively to irrigate gardens regularly but save water at the same time.

The goal of reducing campus water consumption by 25% demonstrates a commitment to sustainable water management. SDG 6 aims to ensure access to clean water and sanitation for all, and reducing water consumption contributes to this goal.

The goal of reducing water consumption and the overall focus on sustainability in campus operations align with SDG 12. This goal promotes sustainable consumption and production patterns, which includes reducing waste and efficiently using resources.

Source 2

Total Water Cost in Manufacturing Industries

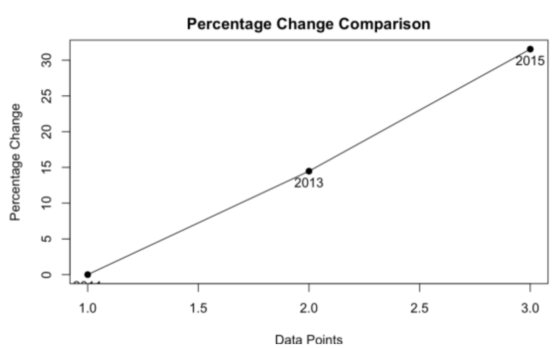
The webpage from Statistics Canada collected the total cost in North American manufacturing industries (*Total Water Costs in Manufacturing Industries, by Water Cost Component and Industry*, 2023). (Note: all the data is x1000 for reference.) The data of 2011 is $1,055,036, 2013 is $1,207,718, and 2015 is $1,387,867. The water cost percentage change could be calculated by

(new data - old data)/old data * 100

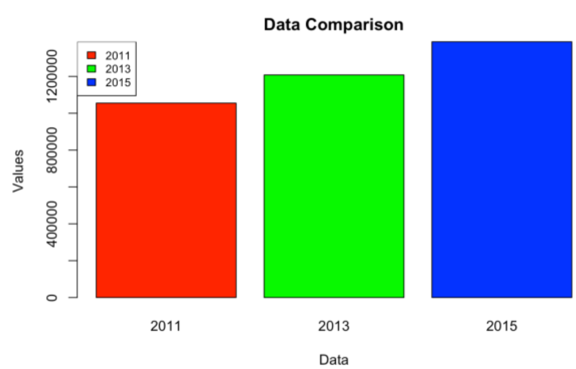
By comparing the year 2011 and 2013, the percentage change formula is ($1,207,718-$1,055,036)/ $1,055,036 * 100 = 14.47%. By comparing the year 2013 and 2015, the percentage change formula is ($1,387,867-$1,207,718)/ $1,207,718 * 100 = 14.92%.

The percentage change from 2011 to 2015 is ($1,387,867-$1,055,036)/$1,055,036 * 100 = 31.55%. The water cost of North American industry is continually increasing as the percentage calculations show. By inputting the data in R language and plotting the percentage change comparison graph (figure1), we can see there exists a positive slope which indicates the positive relationship between these three years' of water cost. As the independent variable of year increases, the dependent variable of water cost tends to increase as well. The larger slope results in a steeper increase of water cost. The coloured bar plot (figure2) gives a direct presentation on how bars are shifting "taller" every two years. As the bars are getting "taller", the water cost is getting "higher". The manufactured greywater systems would help industries save water during the process of manufacturing.

(figure1)



(figure2)



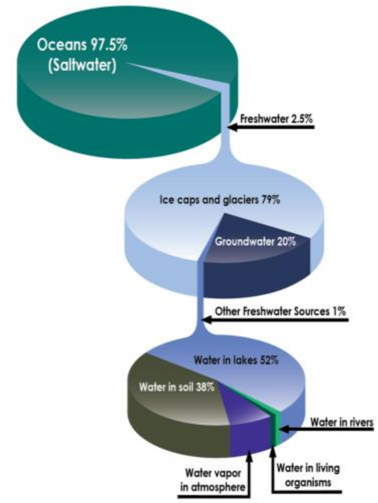

```{r}
industrydata<-c(1055036,1207718,1387867)
qqnorm(industrydata)
qqline(industrydata)
labels <- c("2011", "2013", "2015")
barcolors <- c("red", "green", "blue")
barplot(industrydata,names.arg = labels, xlab = "Data", ylab = "Values", main = "Data
Comparison",col=barcolors)
legend("topleft", legend = labels, fill = rainbow(length(labels)), cex = 0.8)

percentage_change <- ((industrydata - industrydata[1]) / industrydata[1]) * 100
plot(percentage_change, type = "o", pch = 19, xlab = "Data Points", ylab = "Percentage Change",
     main = "Percentage Change Comparison")
text(1:length(industrydata), percentage_change, labels = labels, pos = 1)
```

(original code for generating the plots above)

The increasing water costs in manufacturing industries highlight the need for responsible consumption and production practices. SDG 12 aims to ensure sustainable consumption and production patterns. By adopting water-saving measures and efficient manufacturing processes, industries can contribute to SDG 12's targets.

Source 3

Freshwater and its value!



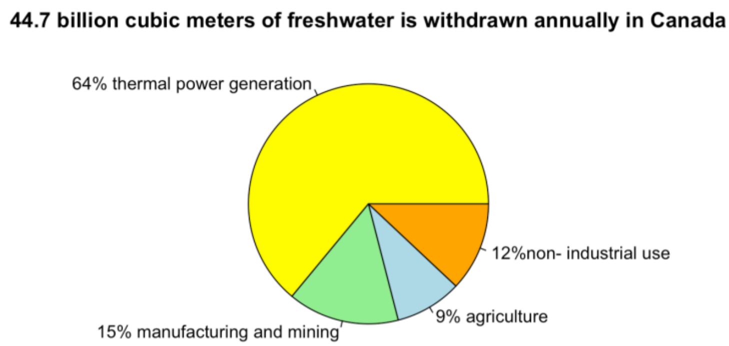
Only 2.5% of the Earth's water is freshwater, of this small percent, a fraction is surface water to be used, the rest is unusable in ice or underground. (Peter H. Gleick, 2018)

0.5% of water on Earth is fresh and accessible. This percent is equal to roughly 8.4 million liters out of trillions and trillions… although this may seem like plenty of water, let's assess the total water usage in CA alone:

411 liters of potable water are used in Canada per person per day (*The Daily — Survey of Drinking Water Plants, 2019*, 2021). There are 40 million individuals in Canada, totalling in over 60 billions in water daily!

How much water is used in Canada?

"Every year Canada withdraws around 44.7 billion cubic meters of freshwater, 64% of which are used for thermal power generation. Manufacturing and mining consume 15%, agriculture 9%, while all non- industrial use (municipal, rural and residential) use only 12%" (*Water Use in Canada | Learn More*, 2022)




```{r}
data <- c(64,15,9,12)
labels <- c("64% thermal power generation", "15% manufacturing and mining","9% agriculture","12%non-industrial use")
color=c("yellow","lightgreen","lightblue","orange")
pie(data, labels = labels,col=color,main="44.7 billion cubic meters of freshwater is withdrawn annually in Canada")
```

(Original code for pie chart above)

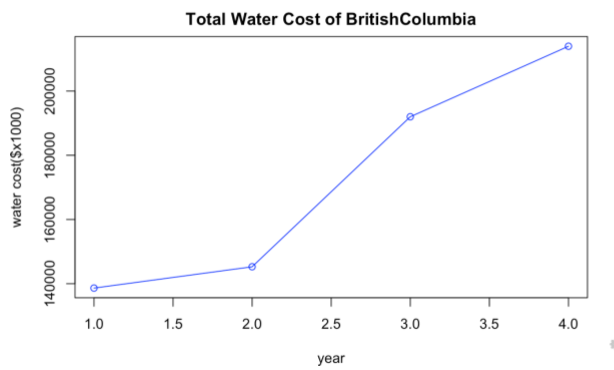
The data and statistics about water usage in Canada highlight the importance of SDG 6, which aims to ensure the availability and sustainable management of water and sanitation for all. The limited availability of freshwater resources and the large quantities of water used in various sectors emphasize the need for efficient water management, conservation efforts, and sustainable practices.

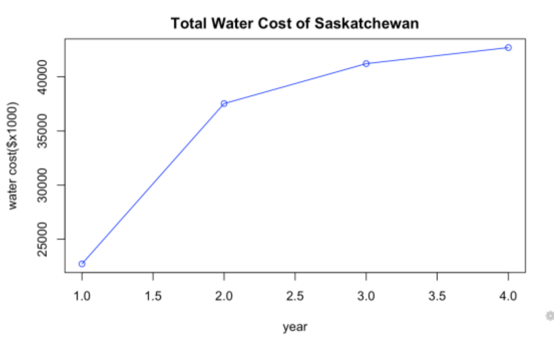
<u>Source 4</u>

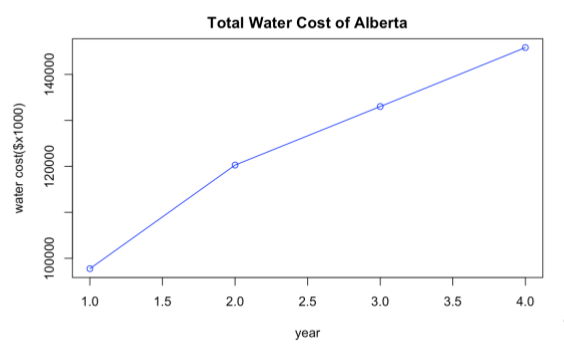
BC Water Usage

From the Statistics Canada website, one webpage gives the specific provisional total water costs in manufacturing industries in different regions. Based on the geographic locations of our business area, there are three provinces chosen for the research to mainly focus on which respectively are British Columbia, Alberta and Saskatchewan (*Total Water Costs in Manufacturing Industries, by Water Cost Component and by Provinces, Territories and Drainage Regions*, n.d.). Those three provinces share most in common depending on region, industrial quantity and acreage.

The growth of total water costs in manufacturing industries (in $x1000) is observed by plotting the line graph of three provinces. From the plots of Alberta and Saskatchewan, both of the two lines are showing a growing water cost but growing slower in gentle slopes. However, the line graph of British Columbia is showing a very steep slope which means the cost of water for manufacturing in British Columbia is continually growing higher.








```{r}
x1<-1:4
BritishColumbia<-c(138632, 145274, 192006, 213976)
x2<-1:4
Alberta<-c(97744, 120281, 133020, 145839)
x3<-1:4
Saskatchewan<-c(22718,  37533,  41223,  42704)

plot(x1, BritishColumbia, type = "o", col = "blue",xlab = "year", ylab = "water cost($x1000)", main =
"Total Water Cost of BritishColumbia")
plot(x2, Alberta, type = "o", col = "blue", xlab = "year", ylab = "water cost($x1000)", main = "Total
Water Cost of Alberta")
plot(x3, Saskatchewan, type = "o", col = "blue", xlab = "year", ylab = "water cost($x1000)", main =
"Total Water Cost of Saskatchewan")
```

(original code for generating the plots above)

The focus on water usage and costs in manufacturing industries directly aligns with SDG 6: Clean Water and Sanitation. By analyzing water costs and trends, it becomes possible to identify areas for improvement in water management, ensuring the availability and sustainable management of water resources.

Source 5

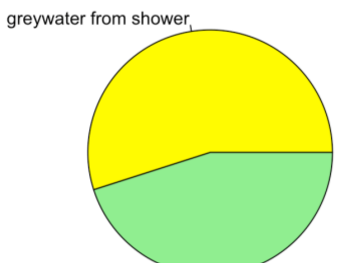
Potential Portable Water Savings

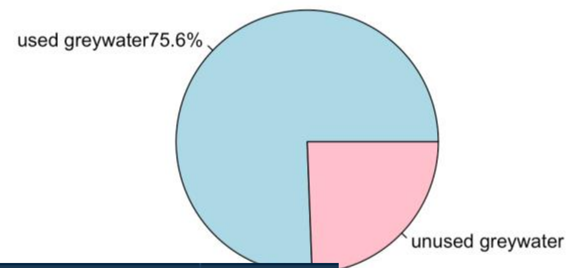
The REUS organization has determined each household occupied by 2.64 people will produce about 193 liters of greywater each day split between 106 liters from showers and 87 liters from clothes washers (Gauley, 2017). By converting the data above into the amount measured by lcd (liters per capita), we use 106 liters from showers/(2.64 person per house) = 40.15lcd, and 87 liters from clothes washers / (2.64 person per house) = 32.95lcd. There is a total of 40.15lcd+32.95lcd=70.1lcd per day.

There were 37 homes sampled for water demand data with an irrigation-based greywater system by the study of Residential Greywater Systems in California (Gauley, 2017). By comparing households without an irrigation-based greywater system, the household with an irrigation-based greywater system was able to save 42 lcd, which means 42lcd/70.1lcd * 100%= 59.9% of greywater is reused each day. We can assume that approximately 60% of greywater was reused right after it got collected.

Furthermore, there are some households have a more efficient greywater usage of 64 lcd greywater used (Gauley, 2017), which means 64 lcd/70.1lcd*100%=91.298% of greywater is used each day. By calculating how much the greywater is reused on average, the formula of counting the average is (59.9%+91.298%)/2=75.6%. Therefore, about 75.6% of greywater is going to be used for housing irrigation.

Assuming the irrigation season is 9 months of 274 days annually, there are 75.6% (actual greywater used) * 70.1 lcd (collected greywater amount)* 274 days=14,520 liters of water would be saved for irrigation.






```{r}

data <- c(40.15,32.95)
labels <- c("greywater from shower", "greywater from cloth washig")
color=c("yellow","lightgreen")
pie(data, labels = labels,col=color)

data <- c(75.6,24.4)
labels <- c("used greywater75.6%", "unused greywater")
color=c("lightblue","pink")
pie(data, labels = labels,col=color)

```

(Original code for pie charts above)

The use of grey water supports SDG 11: Sustainable Cities and Communities: The use of greywater for irrigation contributes to creating more sustainable cities and communities. By reducing the reliance on freshwater sources for non-potable uses, such as watering gardens, communities can achieve greater water resilience and reduce the strain on water resources.

Source 6

UVic: The Bowker Creek Headwaters

The University of Victoria plays an important role as a steward in the operation of Bowker Creek Initiative (*Learn More About UVic's Sustainability Commitments: Uvic.ca/sustainability*, n.d.). Bowker Creek starts at the University of Victoria and flows past Victoria, Saanich and Oak Bay before discharging into the ocean through about 9.4 kilometers for the whole process. The Victoria communities are closely collaborating together in order to maintain the ecology and economy of the Bowker Creek watershed.

The Bowker Creek provides water for various human and agricultural activities, therefore it is important to avoid wasting and polluting it. By installing greywater collection systems on campus, the greywater can be safely stored and transferred to use in a specific pipe. This method helps to reduce the risk of leaking, mixing with Bowker Creek stream which might result in a large range of pollution in the stream.

The greywater system satisfies the SDG 6: Clean Water and Sanitation. The Bowker Creek Headwaters project focuses on safeguarding the Bowker Creek watershed, which provides water for various human and agricultural activities. By implementing greywater collection systems on campus, UVic helps reduce water waste and the risk of pollution, contributing to clean water and sanitation goals.

Source7

UVic: The Seedling Project for Climate and Sustainability Action Plan 2030

The Seedling Project is one of the integrated initiatives of Climate and Sustainability Action Plan 2030 of University of Victoria. As an impact chair in Indigenous Art Practices at UVic, Carey Newman had the idea to plant a western red cedar tree to carve and raise an Indigenous totem after 600 -1,000 years when the tree matures. This initiative represents the Indigenous culture which requires effort and collaboration not only on campus but the community as well.

The greywater collection system can work as a potential support for the future depending on its sustainability. Since the western red cedar tree will require a long time for growth, there might be an economic pressure on maintaining it. Using the greywater collected from the system to irrigate will reduce the cost of gardening.

It is important to note to the companies that once the service is implemented, it works independently and doesn't require extra expenses or upkeep other than regular maintenance like any washroom would. We stress that it will not be of any sort of inconvenience and it is an easy way to make their company more sustainable.

This proves that the greywater system supports SDG 13: Climate Action: By planting a western red cedar tree as part of the Seedling Project, UVic contributes to climate action. Trees play a vital role in carbon sequestration and mitigating climate change.

Implementation

In today's rapidly evolving marketplace, businesses are increasingly recognizing the importance of integrating sustainable practices into their operations. One such solution that holds tremendous potential is the implementation of the greywater collection system.

To implement our greywater system, we must first strategically plan our process. This starts with defining our overall purpose and mission. Our goal is to encourage and create more awareness of resource usage and make changes in our community to be more efficient in our water consumption.

To begin to focus on a target market, we need to identify a consumer group based on research. This includes environmental scanning and surveys to pinpoint our target market. We have identified our target market as businesses with an appreciation for, and have motivation to make sustainable changes to their company. We acknowledge the importance of gathering information about our target customers by first seeing what sustainable actions they've *already* taken to make environmentally positive changes to their businesses. Also can look into what their business is known for, and who *their* market is to use to encourage them to make changes to drive/encourage business for them. If we can convince a company that their consumers will appreciate these changes and therefore engage with their product/service, they will have more motivation to make some environmental adaptations. If a company has made a claim that they are sustainable or eco-friendly, our business can jump on that to put their claim into action, these groups will likely respond favorably and are more likely to be willing to spend money on sustainable services.

It is important as a company to understand our market, the company's demographics, goals, objectives, and their own customers. As well as know their brand personality, how they want to be perceived. Our goal is to encourage businesses to be *proactive*, not *reactive*, make a change before it's too late.

We plan to begin implementation with flyers to distribute within our target market, and to construct an easily accessible and clear website to promote the business. Our plan also includes promotional videos and an interactive website to demonstrate our concepts and help potential customers better understand what the process would be like. To drive website views, we'd send out emails and other online marketing directed towards companies we have researched. This way we can direct them towards the website and share promotional videos of our service, focusing on how it will be beneficial for companies to partner with WaterWise. Once further into the process as an established company, we plan to hire professional consultants to speak directly with companies and set up meetings to discuss potential partnerships.

Current Project of Focus

We are currently focused on the University of Victoria. The University of Victoria has shown their appreciation and already begun to make sustainable changes to their campus and surrounding community. They have proven to be comfortable spending money on sustainable services, and have spoken openly about continuing to do so. To the students and faculty, it is worth it to invest in their future as there is overwhelming respect for the environment from the community of Victoria.

Approaching the University of Victoria:

When approaching UVic, it is important to demonstrate a deep understanding of the university's sustainability goals and challenges. Highlighting the alignment of the solution with UVic's values and objectives will create a sense of shared purpose and make the proposal more compelling. Emphasizing the positive impact the solution will have on the campus and the wider community will further enhance its appeal. Presenting data and evidence supporting the solution's effectiveness will help build trust and confidence.

Summary

Convincing Stakeholders of the Solution's Value:

To convince the stakeholders at UVic of the solution's value, it is crucial to emphasize the long-term benefits. These include cost savings through improved efficiency and resource management, as well as an enhanced reputation and positive brand image. Successful case studies and testimonials from similar institutions can be used to showcase the solution's impact and effectiveness.

By following a well-structured implementation strategy and effectively communicating the value of the sustainable marketing solution, we are confident that it can be successfully implemented at the University of Victoria. The solution's alignment with UVic's sustainability goals and its potential for positive environmental and social impact make it a valuable investment.

In conclusion, the implementation of greywater collection systems presents a viable and sustainable solution for businesses and industries. By capturing and reusing wastewater from various sources, this solution addresses key environmental and economic challenges while contributing to the achievement of several Sustainable Development Goals (SDGs).

Firstly, greywater collection systems promote responsible water management, aligning with SDG 6: Clean Water and Sanitation. By reducing water consumption and conserving natural resources, businesses can contribute to the availability and sustainable management of water resources.

Secondly, this solution supports SDG 12: Responsible Consumption and Production. By implementing greywater collection systems, businesses can reduce their ecological footprint, minimize waste generation, and promote more sustainable practices in water usage.

Thirdly, greywater collection systems contribute to SDG 11: Sustainable Cities and Communities: By implementing sustainability strategies in campus operations, UVic contributes to creating sustainable cities and communities.

Lastly, SDG 13: Climate Action - Efficient water management practices, such as utilizing gray water for irrigation, can help mitigate the impact of climate change by reducing water stress and promoting more sustainable water use.

References

(n.d.). Friends of Bowker Creek Society: About. Retrieved June 22, 2023, from https://bowkercreek.org/

CLIMATE AND SUSTAINABILITY ACTION PLAN 2030 STRATEGY. (n.d.). UNIVERSITY OF VICTORIA. Retrieved June 22, 2023, from https://www.uvic.ca/_assets/docs/csap2030-strategy.pdf

The Daily — Survey of Drinking Water Plants, 2019. (2021, August 17). Statistique Canada. Retrieved June 21, 2023, from https://www150.statcan.gc.ca/n1/daily-quotidien/210817/dq210817c-eng.htm

Gauley, B. (2017, March). *Water Savings & Financial Benefits of Single-Family Package Greywater Systems*. Alliance for Water Efficiency. Retrieved June 22, 2023, from https://www.crd.bc.ca/docs/default-source/water-conservation-pdf/reports/water-savings-greywater-systems-awe-2017.pdf?sfvrsn=ec9a39ca_2

Learn more about UVic's sustainability commitments: uvic.ca/sustainability. (n.d.). CRD. Retrieved June 22, 2023, from https://www.crd.bc.ca/docs/default-source/initiatives-pdf/bci-pdf/interpretive-signs/uvic_bowkercreeksignage.pdf?sfvrsn=1b2185ca_2

Peter H. Gleick. (2018, June 6). *Where is Earth's Water? | U.S. Geological Survey*. USGS.gov. Retrieved June 21, 2023, from https://www.usgs.gov/special-topics/water-science-school/science/where-earths-water

Sustainability Action Plan: Campus Operations 2014-2019. (n.d.). University of Victoria. Retrieved June 22, 2023, from https://www.uvic.ca/sustainability/assets/docs/policy/action-plan-2014.pdf

Total water costs in manufacturing industries, by water cost component and by provinces, territories and drainage regions. (2023, January 16). Statistique Canada. Retrieved June 22, 2023, from https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3810006501

Total water costs in manufacturing industries, by water cost component and industry. (2023, January 16). Statistique Canada. Retrieved June 22, 2023, from https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3810006401

Water. (n.d.). University of Victoria. Retrieved June 22, 2023, from https://www.uvic.ca/sustainability/topics/water/index.php

Water Use in Canada | Learn more. (2022, August 13). Danamark. Retrieved June 21, 2023, from https://danamark.com/resources/water-use-canada/